

Mixed Numbers, Parts, and Wholes

E L A + M A T H • W O R D S T U D Y

Mr. H Math • Day 23

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____

Score: ____ / 12

Ian rides his unicycle **10 yards every 4 seconds**. How long does **25 yards** take? No whole number takes 10 to 25 — the multiplier is $2\frac{1}{2}$. Today you scale ratios by fractions and mixed numbers, and you cross between **part-to-part** and **part-to-whole** the way real recipes, paint jobs, and paychecks do.

DO NOW • SILENT START

3 min

Which two are equivalent? Circle them, then write the number that multiplies one into the other.

A 2 to 3 **B** 12 : 6 **C** 6 : 9 **D** 18 to 9

B → D: multiply by _____

TURN & TALK

2 min

Lemonade: $1\frac{1}{2}$ cups concentrate for every $4\frac{1}{2}$ cups water. How many cups are in one batch? Which ratio is part-to-part and which is part-to-whole?

Frame: "The parts are ____ and ____, so the whole is ____."

SECTION A • GUESS THE WORD

The number you multiply BOTH quantities by to get an equivalent ratio. Fill in the missing letters.

S **C** **L** **E** **F** **C** **T** **R**

Find it by dividing: new amount ÷ original amount = $25 \div 10 =$ _____

SECTION B • WORD BANK

part-to-part **part-to-whole** **scale factor** **mixed number** **cross products**

- Concentrate : water is a _____ ratio. *Compares one part to another part.*
- Concentrate : whole pitcher is a _____ ratio. *Add the parts to find the whole.*
- $2\frac{1}{2}$ is a _____ — a whole number and a fraction together.
- To check whether a : b and c : d are equivalent, compare the _____: $a \times d$ and $b \times c$.

SECTION C • TRY IT: SCALE BY A MIXED NUMBER

WORKED EXAMPLE — Ian's unicycle

Distance (yd)	10	5	25
Time (s)	4	2	10
Move	given	÷ 2	× 5

One step: $25 \div 10 = 2\frac{1}{2}$, and $4 \times 2\frac{1}{2} = 8 + 2 = 10$ s.

Not $8\frac{1}{2}$! Both parts of $2\frac{1}{2}$ multiply the 4.

YOUR TURN — the shampoo

A 32 fl oz bottle costs \$8. A 12 fl oz bottle has the same ratio. What does it cost?

Capacity (fl oz)	32	12
Cost (\$)	8	

Scale factor: _____ Cost: _____

SECTION D • REAL LIFE: CROSS BETWEEN PART-TO-PART AND PART-TO-WHOLE

Situation	Part : part	Part : whole	Scale to...	Find
 Lemonade	$1\frac{1}{2} : 4\frac{1}{2}$	$1\frac{1}{2} : \underline{\hspace{1cm}}$	10-cup pitcher	concentrate = ____
 Concrete	$1 : 2\frac{1}{2}$	$1 : \underline{\hspace{1cm}}$	14 shovels	cement = ____
 Savings	save $2\frac{1}{2}$: spend 10	$2\frac{1}{2} : \underline{\hspace{1cm}}$	\$87½ paycheck	save = ____
 Free throws	made 3 : missed 2	$3 : \underline{\hspace{1cm}}$	40 attempts	made = ____

Cross-product check: Is 8 girls out of 18 students the same girls-to-whole ratio as 12 girls and 15 boys? $12 \times 18 =$ ____ $27 \times 8 =$ ____ Same? ____

REFINE • SUMMARY — Scale factor = new ÷ original, and it can be a whole number, a fraction, or a mixed number. Multiply **both** quantities by it. To cross from part-to-part to part-to-whole, **add** the parts; to cross back, **subtract**. Equal cross products mean the same ratio.

★ Find the multiplier, cross to the whole, check the cross products! ★

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